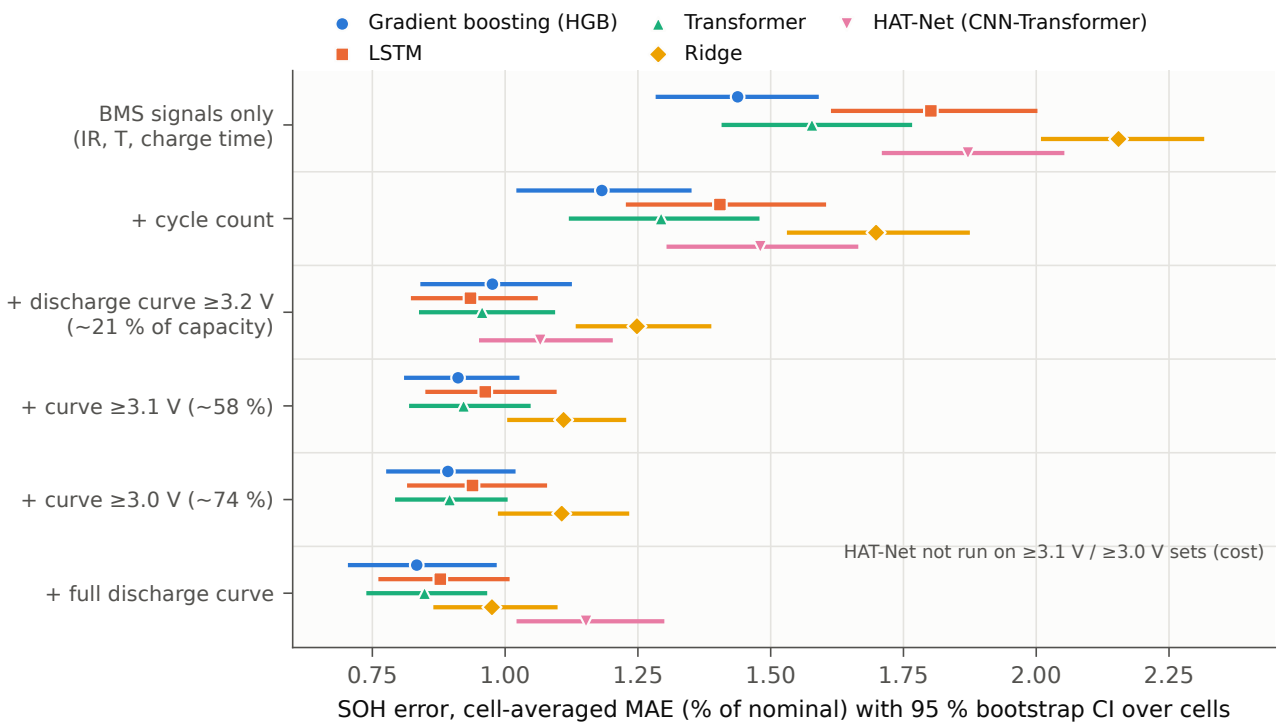
Gradient boosting (HGB)
Transformer
HAT-Net (CNN-Transformer)
LSTM
Ridge
BMS signals only
(IR, T, charge time)
+ cycle count
+ discharge curve ≥3.2 V
(~21 % of capacity)
+ curve ≥3.1 V (~58 %)
+ curve ≥3.0 V (~74 %)
+ full discharge curve
HAT-Net not run on ≥3.1 V / ≥3.0 V sets (cost)
0.75
1.00
1.25
1.50
1.75
2.00
2.25
SOH error, cell-averaged MAE (% of nominal) with 95 % bootstrap CI over cells